

AI Use Case Tool - Red Hat Internship

Title: AI Use Case Discovery Tool

Experience: Global Partner Ecosystem Team, Summer 2026

Role: AI Ecosystem Intern

One-liner: A GTM enablement project bridging the disconnect between partner AI use case marketing claims and real-world field execution through a NotebookLM discovery tool

[Final Presentation](#)

Context

I took on this internship with Red Hat's Global Partner Ecosystem team because I wanted to bridge my Computer Science and Design background at Northeastern with real-world enterprise strategy. Specifically, I wanted to understand how massive open-source tech leaders package complex AI infrastructure and turn dense, multi-partner capabilities into actionable motions that field sellers can actually pitch and close.

The team partnered directly with Red Hat's major hardware (OEMs) and software (ISVs) ecosystem partners, including Dell, Cisco, Lenovo, and Accenture, so I was given the opportunity to directly speak with sales teams, technical reference architectures, and conduct partner interviews. My project was structured in phases: research gathering, schema architecture, database inventory creation, and productizing those insights into a tool for Red Hat sellers.

The Problem

I evaluated two distinct layers of market information: exhaustive partner public marketing claims (websites, press releases) and internal field sales data across enterprise AI deployment patterns (Enterprise RAG, Model-as-a-Service, Agentic AI).

On their own, these were just disconnected decks and unstructured notes. But I identified a major strategic discrepancy: there was a massive gap between the endless AI capabilities partners advertise online and the handful of specific use cases they are actively funding, staffing, and selling in the field.

My theory going in was that Red Hat sellers were wasting critical cycle time pitching joint solutions that partners weren't actually equipped to co-sell in the field, leading to stalled AI deals. Investigating and resolving this was critical to uncover true GTM realities, correct outdated internal assumptions, and align partner infrastructure directly with the Red Hat AI portfolio and the Red Hat AI Factory with NVIDIA.

User & Goals

Persona: The Red Hat Seller

- Day-to-day job depends on credibility during customer discovery calls and identifying actionable joint selling motions without digging through hundreds of page-long pitch decks.
- “When I walk into a meeting with a customer using Cisco or Dell hardware, what is the exact, validated 'next best action' and Red Hat AI attach story I should pitch?”

My Process

Once I validated the gap through interviews with Partner Account Managers (PAMs) and partner leaders, I moved into data structuring. I designed inventory mapping Partner Hierarchy, NVIDIA Focus Areas, Target Vertical, Certified Hardware, Deployment Architectures (RHAIE), and Pre/Post-Sale GTM strategies.

Blueprint Industry: FSI

- To test this framework in a high-stakes, heavily regulated environment, I ran an in-depth blueprint on the **Financial Services Industry (FSI)**.
- I mapped out how complex enterprise scenarios, like Fraud Detection, flow all the way from partner hardware (Cisco UCS, Dell AI Factory) through MLOps platforms (OpenShift AI, vLLM) to the end-user application.

I integrated the structured inventory, case studies, and partner pitch decks into a NotebookLM tool. This turned static field notes into an interactive AI knowledge assistant where sellers could query real-time market realities, obtain pitch scripts, and retrieve technical deployment patterns on the fly.

Implementation: Bringing the Tool into the Real World

A central database is useless if sellers don't adopt it. So the delivery phase became about embedding this knowledge directly into the seller's active workflow. I engineered a bi-directional workflow between NotebookLM and the core AI Use Case Inventory.

- A seller preparing for an account call can ask NotebookLM questions like: *"What regulatory challenges prevent banking customers from deploying Gen AI for fraud detection?"* or *"What is the 'next best action' for a client on Cisco hardware?"*
- The tool instantly outputs a tailored talk track alongside structured, copy-pasteable database rows. If the seller uncovers a new use case in the field, they copy that row directly back into the master inventory.
- This ensures the database acts as a living, self-updating asset for the entire global GTM organization.

Data Visualization - ARTG2242

Title: Open Space & 311 Requests: Visualizing Equity in Boston's Public Services

Course: ARTG2242, Information Design Principles, Spring 2026

Role: Data Visualization Designer + Researcher

One-liner: A data visualization project examining the relationship between open space access and city service responsiveness across Boston neighborhoods.

[Final Presentation](#)

Context

I took this course in spring 2026 because I wanted to get better at data visualization, specifically how to take dense civic datasets and turn them into something the public could actually read and act on. I was also looking to get hands-on with tools I hadn't used before, like Tableau and Rowboat, rather than just studying data viz in the abstract.

The course partnered with Analyze Boston, so we worked directly with real city data instead of sample sets. Each of us picked a dataset and worked through it in phases: exploration, data cleaning, then building visualizations for different audiences. Some visualizations were built for professionals close to the dataset, others for everyday residents. The goal was to understand how the same data can serve very different purposes depending on the audience: communication, awareness, analysis. This project became my testing ground for that whole process, working through Boston's own city data.

The Problem

I chose two Boston city datasets. The Open Space dataset maps every publicly designated park, playground, and protected green space in the city. The 311 Request dataset is a continuously updated log of every service request residents file, such as potholes, missed trash pickups, and broken streetlights, tracked by what was requested, where, when, and whether it got resolved.

On their own, these are just civic records. But I had a hunch they were connected to a bigger story. My theory going in was that wealthier neighborhoods get to reinvest in the parks that made them desirable in the first place, creating a self-reinforcing cycle, while under-resourced neighborhoods never get pulled into that same loop. I wanted to see whether the data backed that up, and whether city responsiveness followed the same pattern.

User & Goals

Before I touched a single chart, I built a persona to stay grounded in who this was for: Lindsey Matthews, an urban planner. Her job depends on two things: making sure residents have quality open space within walking distance, and building public trust through transparent planning. Her real question was simple: is the distribution of resources getting better or worse over time, and where should attention and funding go next? That question shaped almost every decision I made after this point.

Process: Building the Dashboard

Once I had the datasets and the guiding question, I moved into Tableau. I wanted someone to understand the state of the city in under a minute, so I led with three KPI cards: 50,265 total 311 requests, Public Works as the most common responding department, and 70.5% of requests resolved on time. Those numbers told a story of a city that's mostly functional, but not everywhere.

From there, I built three more views. A choropleth map shows acreage of open space by neighborhood, so you can see where the green space actually is. A scatter plot compares open space per capita against 311 requests per capita, neighborhood by neighborhood, and this turned out to be the most important chart in the project, since the line of best fit came out almost flat, with a slope of roughly $+0.04$. In plain terms, more open space doesn't mean fewer service complaints. Dorchester had the highest volume of 311 requests in the entire dataset, at 1,435. Two bar charts sit side by side, one showing total open space per neighborhood and the other showing total 311 requests per neighborhood, so you can compare both without holding numbers in your head.

Key Insights and Learnings (both from the data and the process)

That flat scatter plot trendline became the core finding of the project. I expected open space and service responsiveness to move together in some way, but instead, the data suggested some neighborhoods are underserved on both fronts at once. That's a more troubling story than the one I started with, and it changed how I approached the next phase.

Getting to that insight also taught me something about working with real civic data instead of clean sample sets. Messy records and inconsistent naming meant I had to slow down and verify things I would have otherwise assumed were reliable. The fact that the 311 system was mid-restructuring was a reminder that public data is often a snapshot of a moving target, not a finished record, and that any conclusion I drew needed to account for what the dataset couldn't yet tell me. That shaped how carefully I worded the final insights, since I wanted to represent the pattern honestly without overstating certainty the data didn't support.

Implementation: Bringing the visualization into the real world

The dashboard was useful, but no resident would ever open it on their own. So the communicating phase became about putting the same insight into something that could live in the real world. I designed a series of posters for public parks, bus stops, and other high-traffic spots, each one carrying a QR code linking to more information, resources, and contact info. I thought about who would actually stop and read one: residents curious about their own neighborhood's 311 responsiveness, people frustrated enough to want a clear call to action, and new residents comparing neighborhoods before they move.

Each poster is built around one of three ratios: requests to open space acres, requests to overdue requests, and requests to closure rates, and each one is localized to a specific neighborhood using its own numbers.

Roslindale's poster leads with "156 Wins, More Possible." Dorchester's reads "Asks the Most, Gets the Least," a direct nod to what the scatter plot showed. Jamaica Plain's is blunter: "122 Ignored." I also pulled in citywide 311 complaint data to give the posters more weight. Parking enforcement was the single largest complaint category, at over 63,000 requests, followed by street cleaning, improper trash storage, and pothole repairs. Under-resourced neighborhoods carry a heavier share of these requests overall, which reinforces the same pattern the dashboard surfaced.

AI Literacy Curriculum - Exploration

Title: AI Literacy Curriculum

Course: ARTG5000, Designing AI for Education, Spring 2026

Role: Curriculum Designer + Researcher (co-designed with a class partner)

One-liner: A modular AI literacy curriculum for grades 5 to 8 that teaches students to recognize hallucinations, bias, and sycophancy through hands-on activities.

[Final Presentation](#)

[AI Literacy Curriculum](#)

Context

This course, Designing AI for Education, was one of my favorites at Northeastern University. The final project was open-ended, as long as it fell into that broad theme. My partner and I built this curriculum because we noticed a gap in how AI gets taught to kids. Middle schoolers already use AI every day to write essays, answer questions, generate images, and get recommendations, but almost none of them have been taught to question it.

Most of what already exists falls into one of two traps. It either teaches kids how to use specific tools that will be outdated within a year, or it goes so deep into ethics and systems theory that it loses the average 7th grader in the first 10 minutes. We wanted to build something in between that actually sticks. We wanted to convey one simple message: AI is not magic, and it's not always right. It's a tool that's only as good as the data it was trained on and the way you interact with it. Middle schoolers are old enough to grasp that, and they're exactly the right age to start pushing back on technology instead of just consuming it.

The Problem

We were designing for two audiences at once. The primary audience is middle school students in grades 5 through 8. This encompassed kids who are skeptical enough to question things but still open enough to engage with new ideas. They're on YouTube, TikTok, and Google all day, so they're already living inside AI systems whether anyone teaches them to think about it or not. The secondary audience is the teachers who deliver the content, most of whom have no technical background and need something they can run without extra prep.

We then did some research on what already exists. A well-known curriculum from MIT Media Lab was one we used as inspiration and a starting point. However, this curriculum framed AI mainly as a socio-political artifact: who built it, who it harms, what power structures it reflects, with heavy emphasis on recommendation algorithms and facial recognition. That framing matters, but we felt that it was outdated. Students don't know how to use AI well, and they don't know why it fails them when it does. We wanted to build the practical foundation first, so our version asks a different question: how do I use AI well, and why shouldn't I blindly trust it? Where MIT's curriculum treats AI as a mirror of society, ours treats it as a malleable tool, and we made skepticism the explicit outcome we were designing toward.

User & Goals

We wanted students to leave with three things.

- First, genuine skepticism, not as an abstract fact but as something they'd seen happen with their own eyes: AI gives confident-sounding answers that are sometimes completely wrong, and it generates images that reflect stereotypes because it was trained on data that wasn't neutral.
- Second, a sense of control. One thing that bothers us about how AI gets talked about everywhere, in the news, in schools, is this sense of inevitability, like AI is just happening to us and there's nothing to do but adapt. We wanted students to understand that AI responds to how you prompt it, that it was built by people who made deliberate choices.
- Third, the confidence to push back, not on AI specifically, but on anything that presents itself as authoritative without earning it.

Process: Building the Curriculum

We designed three modular activities that teachers can run independently, each built around a single class period and requiring no technical background to facilitate.

1. **Descriptive Prompting:** Students write step-by-step instructions for making instant ramen, which the teacher then follows hyper-literally in front of the class. Telling someone to "boil the water" causes the teacher to freeze, and "add the seasoning" gets the sealed packet dropped straight into the cup. The breakdown makes the point before we ever say it out loud: vague instructions produce useless results, and that's exactly how AI prompting works too.
2. **Identifying Bias:** Asks students to think of a real person they know in a specific professional role, a doctor, a teacher, an engineer, and then prompt an AI image generator with just the job title. Comparing what the AI generated against the real person they had in mind makes bias feel concrete instead of abstract, since it's not a stranger's face being flattened into a stereotype; it's someone they actually know.
3. **Fact-Checking AI:** Uses two secretly different chatbots we built in Playlab. One is designed to always sound confident even when it's wrong; the other is designed to always agree with the user no matter what they say. Students interrogate both bots with the same set of questions and log what they notice, working like detectives to figure out what's different about each one, before we ever introduce the terms hallucination or sycophancy. Once they've named the pattern themselves, they get to rewrite the bot's instructions and test whether their fix actually made it more honest.

Alongside the three activities, we also built a lightweight prototype tool where a teacher can input their grade range, available time, classroom tech access, class size, and students' prior exposure to AI, and get a recommended activity pulled from the curriculum automatically.

Key Insights, Learnings, and User Feedback (Teacher Interviews)

Running this by real teachers taught us as much as building it did.

- **Michael Kim**, a math teacher who works in a 1:1 setting, said the personal connection in the bias activity is what makes it land; comparing an AI output to someone you actually know makes the gap feel real instead of distant, but he flagged that the discussion questions needed adjusting for a one-on-one format rather than a full classroom.
- **Marlene Goncalves**, a 7th grade English teacher, connected the fact-checking activity directly to something she already sees in her own classroom: students treating Google's AI summary as fact without questioning it. She also pushed back on something we hadn't fully accounted for, which is that the bias activity opens a conversation about race and gender that not every teacher will feel equipped to handle without explicit ground rules and vocabulary front-loaded beforehand.

That feedback reshaped how we think about the curriculum's boundaries. We built this to teach skepticism toward AI specifically, but a few of the discussions it opens are bigger than AI, and the curriculum needs to hand teachers the scaffolding to hold that conversation responsibly rather than assuming they'll improvise it. We also learned that pacing is far less predictable than we assumed. The same activity can run twelve minutes or forty-five minutes depending on how deep a class wants to go, which pushed us to think about extension prompts for classes that move faster than expected, not just support for classes that need more time.

***ADD A CONCLUSION (?)**

BHCHP - Product Design

Title: Volunteer Application Portal for Boston for Boston Health Care for the Homeless Program

Experience: Code4Community, Northeastern University Club, Fall '25 - Spring '26

Role: Design Lead (team of 9)

One-liner: An application portal for BHCHP to better track their volunteers applications in a functional and sustainable way.

Context

The Problem

User & Goals

Process: Building the Dashboard

Key Insights and Learnings (both from the data and the process)

Implementation: Bringing the visualization into the real world

Eddy - Product Design

Title: Eddy: Fostering Interactive Learning in the Classroom

Course: ARTG 2400, Interaction Design Principles, Spring 2025

Role: Product/UX Designer, team of 4

One-liner: A classroom platform that pairs elementary students across different schools to close equity gaps in access to technology and enrichment.

Context

This project came out of an interaction design course where our assignment was to design a product around one of the United Nations Sustainable Development Goals.

Our team chose Quality Education, which is about ensuring inclusive and equitable education and promoting lifelong learning opportunities for everyone. We wanted to take that broad goal and turn it into something specific and buildable, so we started by figuring out what inequity in education actually looks like on the ground, not just in theory.

The Problem

We started with our own experiences. Each of us went to a different public school growing up, and comparing notes made the gap obvious immediately.

- Jayden's school in Central New Jersey was well funded with art, music, and computer skills programs.
- Saanvi's school in Acton, Massachusetts had music, STEM, and debate programs, plus a district wide one to one device program that started in 2020.
- My school in Brookline, Massachusetts was well funded with some extra STEM and arts programs.
- Olivia's elementary school in Brooklyn, New York had limited STEM programs and wasn't especially diverse. Just within our own group, funding and access varied a lot, and none of us had gone to a school anyone would call under-resourced.

So we went looking for a comparison point outside our own experience.

- We interviewed Will Newkirk, who attended a public K through 8 school in inner city Minneapolis. His school was underfunded through the 2010s due to the recession and other factors, and he never had access to personal computers to practice typing or basic technology skills.
- He told us he still prefers pencil and paper over computers, and that many of his peers are now in blue collar jobs with little to no experience working with a computer. That interview made the stakes of this project concrete in a way our own experiences hadn't.

We also talked to Saanvi's mom, who teaches elementary school in Boxborough, Massachusetts.

- She told us that technology integration varies widely based on individual student needs. Some kids benefit a lot from tech, others get distracted or overstimulated by it.

- She said the ideal solution would involve parent-inclusive tech tools and cross-school collaboration for special needs education, and noted that while kids in her district tend to be fairly tech proficient because of the area's higher income levels, it really comes down to the home environment and how involved a student's parents are.

On the market side, we found that the digital education market is expected to grow at a compound annual rate of around 28 percent, reaching an estimated 66.7 billion dollars by 2028, with North America holding the largest share of that market. Existing tools like ClassDojo and Discovery Education charge around 109 dollars a year per subscription, and self-paced learning is growing even faster than instructor-led formats, which told us there was real room for something new, but also real competition already in the space.

All of this pointed us to one guiding question: **how might we create equitable learning opportunities for elementary school students across different income levels in Boston by creating technology that encourages personalized learning?**

Ideation

Once we had a direction, we split into small brainstorms and pulled ideas together. Some of the early concepts included:

- A video conferencing interface for real-time collaboration on projects
- A digital library of educational activities that also worked offline for students without reliable internet
- Gamified prompts to keep engagement up
- Structured activities designed to help kids get to know an assigned virtual buddy.

We looked at existing kid-facing platforms like Zoom, Google Meet, and PBS Kids for interface inspiration, since we wanted something that felt playful and familiar rather than like enterprise classroom software.

From that brainstorm, we landed on five core features: a buddy pairing system built on an interest survey, a variety of educational activities, customizable student profiles, gamification elements like points and levels, and a class roadmap so both students and teachers could track progress.

We also spent real time on branding, since this needed to feel approachable to a seven or eight year old, not like a tool built for adults. We named the product Eddy, combining the beginning of the word education with a mascot we could personify. We tested a handful of animal mascots, a fox, bear, frog, rabbit, cat, husky, and panda, before landing on the fox, and built out a full type system, button styles, and color palette to keep the interface consistent and kid-friendly throughout.

Prototype

- Show demo from presentation

Implementation: Bringing It Into the Real World

Once we had a working prototype, we thought hard about how Eddy would actually function inside a real classroom, and that meant designing for very different levels of school resourcing.

For schools with one to one device programs, which currently make up only about half of school districts in the U.S., Eddy would run on each student's own device. For schools without that kind of access, we designed around what's actually available: shared computer labs, students partnering up on a single device, or a teacher running Eddy through an overhead projector or interactive whiteboard for the whole class at once.

We also mapped out how a typical session would work. A teacher registers their class and a partner class, introduces the program to students, and prepares learning prompts ahead of time. Students take a short interest survey, get matched with a buddy from the partner school, and then work together during a designated 30 to 45 minute Eddy time block that the teacher controls.

Safety was something we had to design around directly, given how much concern exists around kids and unsupervised internet access. Our answer was to keep Eddy fully contained: access is limited to the classroom and school hours only, accounts are distributed exclusively by teachers, and there's no path for outside contact from anyone outside the paired classrooms.

We think the potential impact goes beyond just tech access. Students in more homogenous or monocultural areas would get to interact with kids from a different district in a monitored, equal setting, building social skills around people who are different from them, being exposed to interests or hobbies that might not be encouraged in their own community, and building stronger digital communication skills along the way.

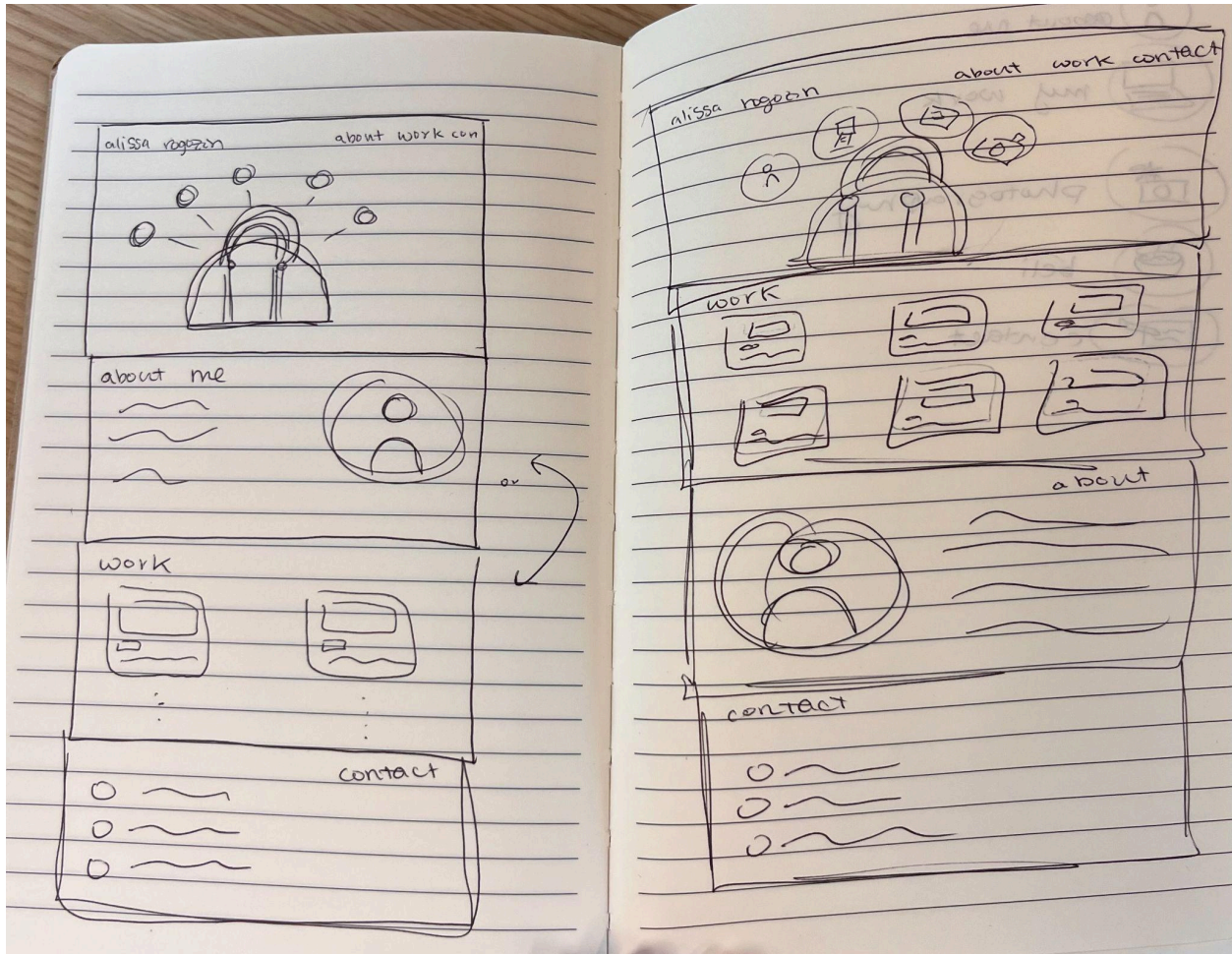
Reflection & Learnings

Designing around a Sustainable Development Goal turned out to be harder than we expected, mainly because inclusivity and integration aren't things you can solve with a single feature. We went through a lot of iterations and a lot of ideas before we landed on the buddy system as the core mechanic, and even then we know it's a partial answer to a much bigger problem. Looking ahead, we'd want to expand Eddy beyond a single city, add more features for parent involvement and collaboration, give students a path to eventually meet their buddies in person, run actual user testing with elementary school students instead of relying on our own past experience, and integrate self-paced learning modules to match where the broader market is heading.

Sources

- [Class.com](#)
- ClassDojo
- Discovery Education

Inspiration and Wireframes



Inspiration Portfolios:

- <https://nabauer.com/>
- <https://www.glorialo.design/>
- https://uxfol.io/d4cda5a1?utm_source=blog&utm_medium=header-link&utm_campaign=examples
- https://uxfol.io/48c83e26?utm_source=blog&utm_medium=header-link&utm_campaign=examples